

# DIFFERENCES-IN-DIFFERENCES ESTIMATION OF INDEX INCLUSION EFFECTS

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*We examine the effects of S&P 500 index inclusion from 2007 to 2023 using a matched, event-study differences-in-differences specification. We find evidence for the lack of any trading price abnormalities but noticeable effects on volatility and especially volume and offer explanations for these results in the context of existing literature. We consider directions for future research as more pertinent and sophisticated econometric methods develop.*

## 1 Introduction

The S&P 500 is the United States’ foremost stock index and, indeed, the world’s most important; membership in it qualifies a company as one of its industry’s domestic and global leaders, and the date of accession is seen as a monumental corporate milestone. In addition to its combined market capitalization of roughly \$70 trillion (as of October 2026), financial instruments worth tens of trillions more depend directly or indirectly on the index’s performance. It is no surprise, then, that the actual event of index inclusion receives significant attention from traders, financial analysts and academics alike looking for opportunities to exploit or phenomena to study.

From a theoretical standpoint, the efficient-market hypothesis introduced by Fama (1970)—specifically the semi-strong and strong versions—imply the immediate “pricing-in” of all publicly available information, meaning that index inclusion should not generate any unusual trading behavior. However, Shleifer (1986) found positive abnormal returns on the announcement date of inclusion that persist for at least ten days, which he attributes to buying by index funds (the most common and obvious explanation even today, since they of course need to balance their holdings); Harris and Gurel (1986) corroborate with similar results, which they see as evidence for price pressures. Since then, a substantial literature that studies both U.S. and international markets using various econometric methods has developed; a survey can be found in Afego (2017).

Relatedly, significant advancements have been made in the area of event-study differences-in-difference (DiD) estimation over the past few years, with several articles coming out that show the bias and/or inadequacy of the traditional two-way fixed-effects (TWFE) estimator and offer new regression specifications

in response. This paper employs a relatively strong matching assumption (that we believe is justified in this instance) to sidestep the potential issues in previous studies, but we expect future work to use these more novel methods, and as such, we offer a brief discussion toward the end. Of course, our main results remain our estimations of dynamic treatment effects for log returns, log volume change and log volatility around the inclusion date: we find no discernible changes in returns and an obvious pattern for volume, with volatility being subtler though still clearly significant.

The rest of this paper is structured as follows. Section 2 offers reviews of both the index inclusion effect and event-study DiD literatures. Section 3 discusses the empirical and econometric methodology used. Section 4 presents regression estimates. Section 5 comments on the results and gives suggestions for future research. Section 6 concludes.

## 2 Literature Review

Since the initial pair of papers in 1986, most investigations have continued to focus on price effects as the main subject of study. An interesting and relatively early example is Chen, Noronha and Singal (2004), which observes an asymmetry between additions to and deletions from the S&P 500: they hypothesize that when a stock is added, increased investor awareness contributes to the observed permanent price increase, but when a stock is removed, it does not suffer the reverse effect to the same degree. Internationally, Yun and Kim (2010) examine the South Korean KOSPI 200 and also find long-term price effects with limited reversal; in addition, they document persistent increases in trading volume, in part due to anticipatory trading. Liu (2011) observes a similar permanent price effect for the Japanese Nikkei 225 but attributes it to lower post-inclusion volatility, which is the opposite of findings from the S&P 500. Indeed, turning attention to measures of liquidity (i.e., volume and volatility), Hegde and McDermott (2003) outlines a mechanism of decreased transaction costs and information asymmetries when a stock is added to the S&P 500 (and vice versa). Building on this, Lin (2007) shows clear increases in both stocks' trading volume and volatility upon inclusion, in large part due to the influence of index-based futures and options. More recently, Chang, Hong and Liskovich (2015) employ a regression discontinuity (RD) design on the Russell 1000/2000 cutoff to give direct causal evidence that supports much of the above; the matching regime that we use (explained in the next section) is of a similar spirit.

However, as index inclusion has been studied more and more, it seems to have also evolved in a possibly self-perpetuating cycle: a significant development came just last year in Greenwood and Sammon (2025), which finds that the S&P 500 inclusion effect on price has fallen from 7.4% in the 1990s to less than 1% since 2015 (and similarly stark contrasts for some other comparisons). It seems that over the past few years, the scholarly consensus has only grown sparser, and while positive price and volume effects still seem relatively entrenched, volatility might well be a toss-up, especially when considering non-U.S. equities. As such, there is more need than ever for pertinent econometrics to ascertain accurate causal

effects, and luckily, the early 2020s provided four innovative papers in the field of event-study DiD: Goodman-Bacon (2021), Callaway and Sant’Anna (2021), Sun and Abraham (2021) and de Chaisemartin and d’Haultfœuille (2023). We delay a thorough discussion of their results and applicability to Section 5, but as an introduction, a natural DiD regression to run would be on panel data of all stocks added to a certain index over a (long) period of time. Over several years, a stock would start as a not-yet-treated control, then be added and finally become part of the treated group (and possibly even cycle in and out of the index multiple times); the entirety would be aggregated to estimate a single average treatment effect comparing before versus after index inclusion—the classical TWFE. However, the subtleties of interacting calendar and event times, as well as treated and control groups changing over time, generates unexpected problems; the four papers mentioned above identify these issues and/or offer solutions to them.

### 3 Empirical Methodology

We obtained data on 191 S&P 500 inclusions from 2007 to 2023. (All data used in this paper is available on the author’s website at <https://github.com/aster-yz-li/index-inclusion>.) Surprisingly, there is no public repository of S&P 500 component changes that is both comprehensive and authoritative, so we manually compiled our list using S&P Global news releases, corporate announcements and other reliable journalistic sources. (We do not claim full coverage but believe the omissions to be sufficiently sparse and unbiased.) We used the Python package `yfinance` to download daily price and volume data for approximately one calendar year before and after each stock’s inclusion date. Companies that changed tickers or underwent M&A events after inclusion were carefully handled to ensure accuracy, though a small number had to be discarded due to data unavailability, usually due to delisting. Two stocks in our sample entered the index twice (having dropped out in between) during the time period considered; for these cases, we only retained the latter instances.

To create a control group, each treated stock was matched with a stock from the S&P 400 (the mid-cap index sitting just below the S&P 500) from the same GICS sector; specifically, every control stock was in the S&P 400 (according to Bloomberg Terminal records) in the same year its corresponding treated stock entered the S&P 500. In case of multiple suitable controls, the one with the closest market capitalization (almost always largest) to the treated stock was chosen, and as a precaution, no stock in the treatment group was ever used as a control, even if the two inclusion dates would have been far apart. This RD-style matching (recall Chang, Hong and Liskovich (2015)) within sectors—designed to control for sector- and market-wide changes in aggregate, thus suppressing confounding with, e.g., macroeconomic events—provides strong *ex ante* evidence of pre-treatment parallel trends, which is the central assumption of our identification strategy. Our data panel is aligned by event date from  $-250$  to  $250$  trading days, with the inclusion date designated as  $t = 0$ .

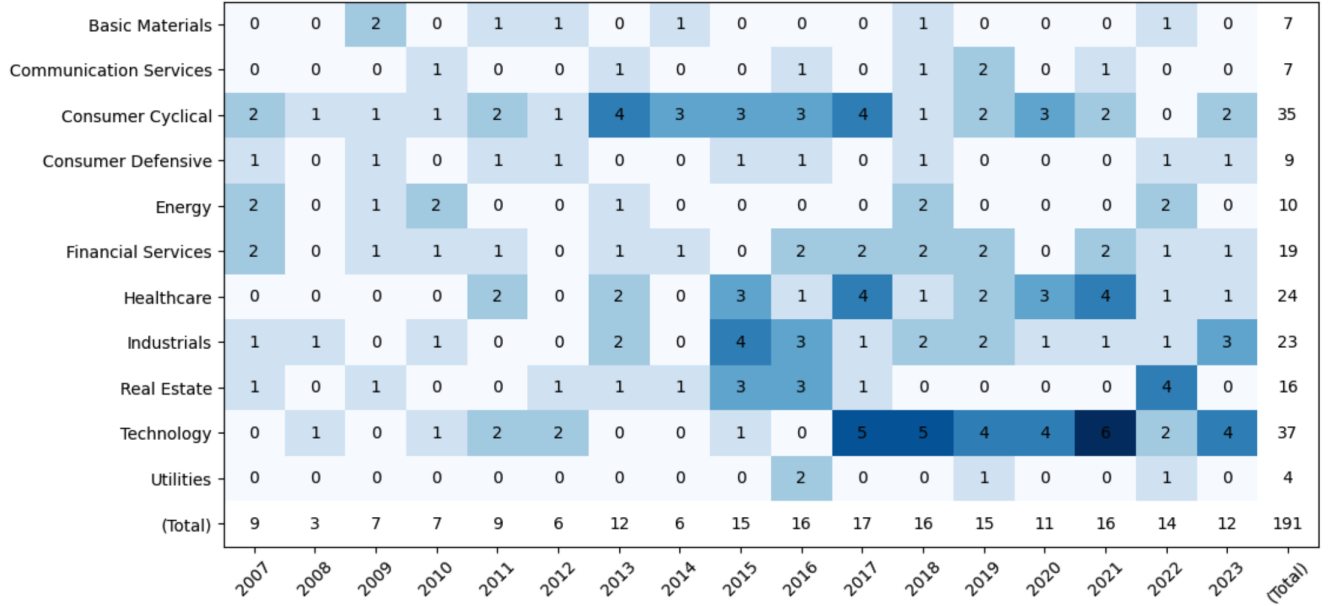


Figure 1: Heatmap of inclusions by sector and year.

We consider three dependent variables: log return, log volume change and log volatility, defined for stock  $i$  on event-(trading-)day  $t$  as follows, where log is the natural logarithm,  $p_t$  is the adjusted closing price and  $v_t$  is the trading volume.

$$\logreturn_t = \log \left( \frac{p_t}{p_{t-1}} \right)$$

$$\logvolumechange_t = \log \left( \frac{v_t}{v_{t-1}} \right)$$

$$\logvolatility_t = \log \left( \sqrt{\frac{252}{20} \sum_{s=t-10}^{t+10} (\logreturn_s - \overline{\logreturn_s})^2} \right)$$

The return and volume change measures are standard, but the volatility measure requires some explanation. We use the logarithm of the annualized standard deviation of daily log returns over a 21-day window centered on day  $t$ . (The numerator of 252 is for annualization, while the denominator of  $20 = 21 - 1$  reflects the degree of freedom adjustment). The 21-day window approximates one trading month and smooths the noise of daily returns, though this does introduce high serial correlation (which we address in Sections 4 and 5 by clustering standard errors at the matched-pair level and consider when interpreting results). Finally, the logarithmic transformations place all three dependent variables on convenient proportional scales and reduce the influence of extreme values.

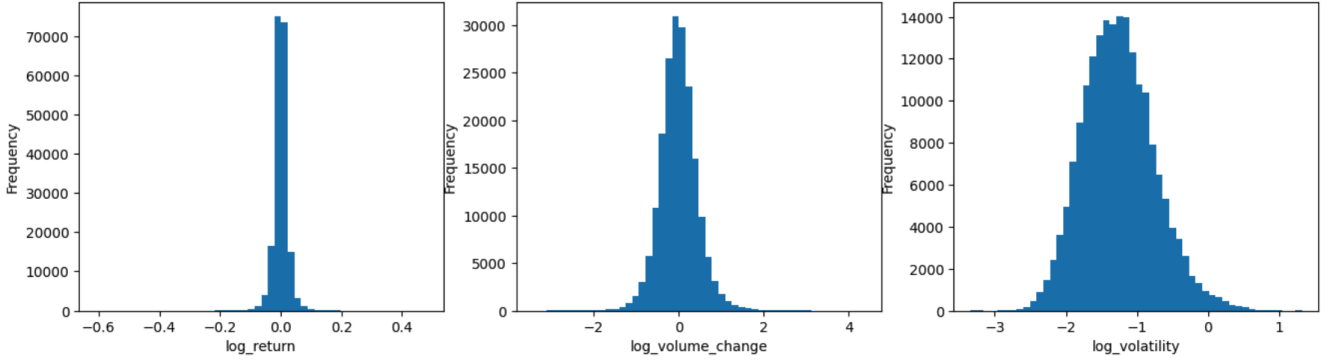


Figure 2: Histograms of dependent variables across panel ( $n = 191382 = 191 \times 2 \times 501$ ).

We now turn to our regression specification. Define

$$\Delta Y_{p,t} = Y_{p,t}^T - Y_{p,t}^C,$$

where  $Y$  denotes one of the three dependent variables,  $p \in \{1, \dots, 191\}$  is an index over the matched pairs,  $t \in \{-250, \dots, 250\}$  is an index over event time,  $T$  indicates the treated stock in the pair and  $C$  indicates the control stock in the pair. We then estimate

$$\Delta Y_{p,t} = \alpha_p + \sum_{k=-250, k \neq -100}^{250} \beta_k \mathbf{1}\{t = k\} + \epsilon_{p,t},$$

where  $\alpha_p$  are pair fixed effects,  $\beta_k$  are our coefficients of interests (time-dependent effects of index inclusion),  $\mathbf{1}\{t = k\}$  are indicator variables, and  $\epsilon_{p,t}$  is idiosyncratic error. We omit the indicator for  $t = -100$  to set it as our “baseline”: all  $\beta_k$ ’s are measured as differences from it. We choose this somewhat arbitrary date because it is far enough from treatment to not suffer from any possible anticipation effects; our results are robust to changes in the reference period (also evident by inspection of the plots in the following section). Due to difficulty in obtaining data, we do not add any covariates (the primary one that we considered is market capitalization, but its heavy right-side skew makes it unsuitable in our judgement).

Before presenting our regression results, we briefly address the nature of treatment itself in the context of this study. Our inclusion dates are when stocks officially join the S&P 500 (i.e., when they become components in its calculation), but *announcements* of inclusion usually occur several weeks prior. We do not use the latter due to its inconsistency (announcement-to-inclusion periods are not always the same exact amount of time) and the former’s greater technical importance for, say, derivatives trading and index fund rebalancing. In any case, our time-dependent  $\beta_k$ ’s (in contrast to the single coefficient of a TWFE regression) allow us to interpret pre-treatment effects as possible post-announcement anticipation.

## 4 Regression Estimates

Figures 3-7 show  $\beta_t$  estimates for different dependent variables; 95% confidence intervals are shaded.

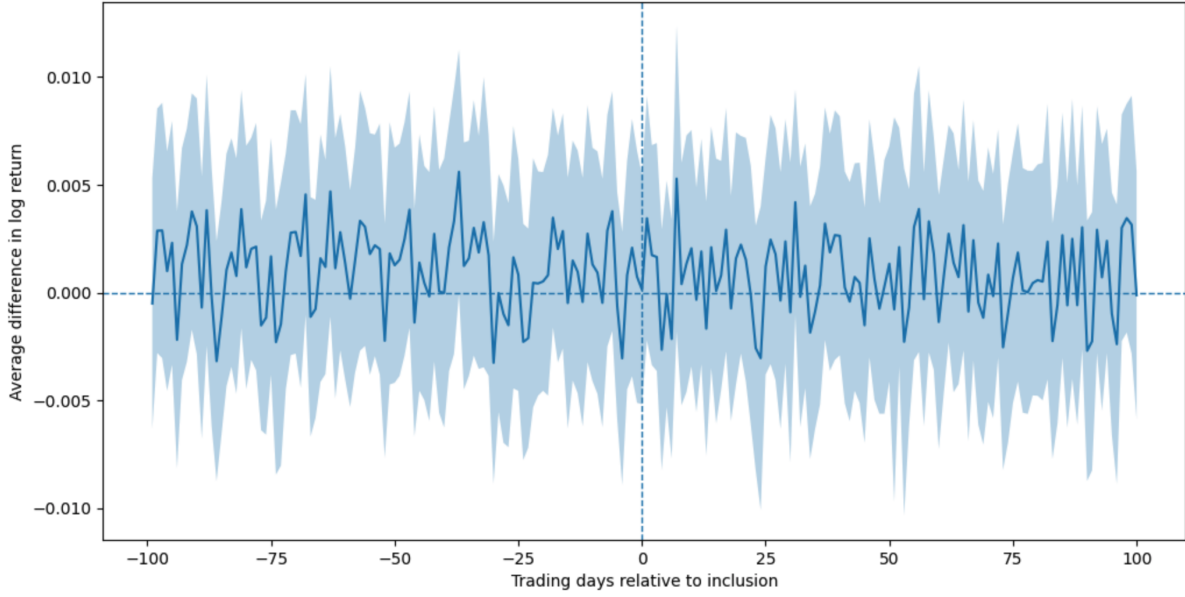


Figure 3: Log return;  $[-100, 100]$  window shown.

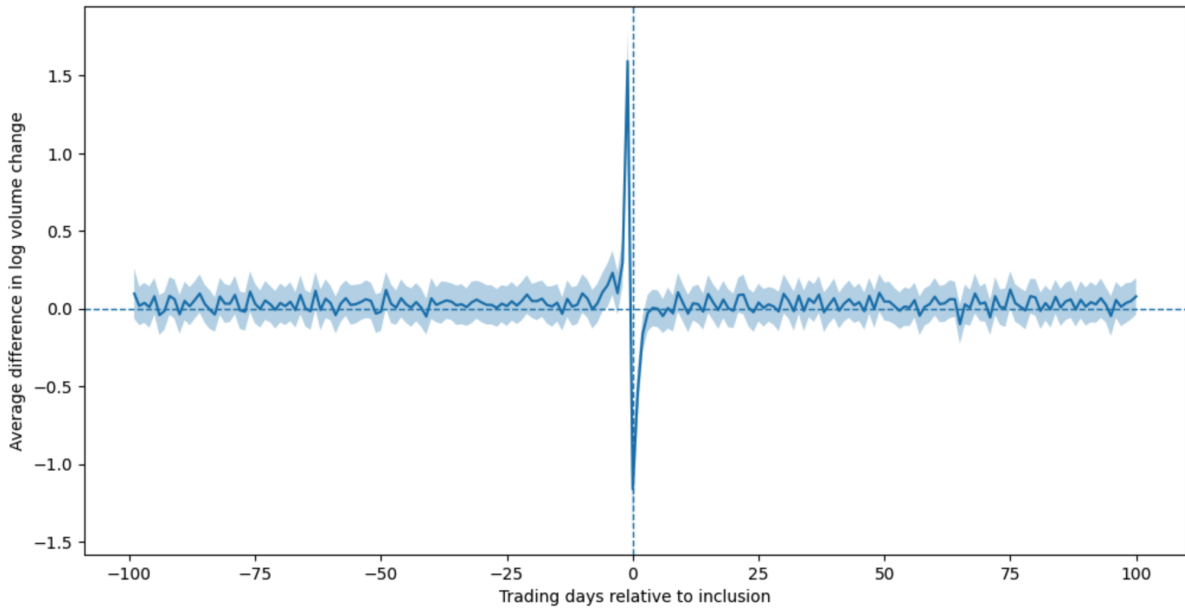


Figure 4: Log volume change;  $[-100, 100]$  window shown.

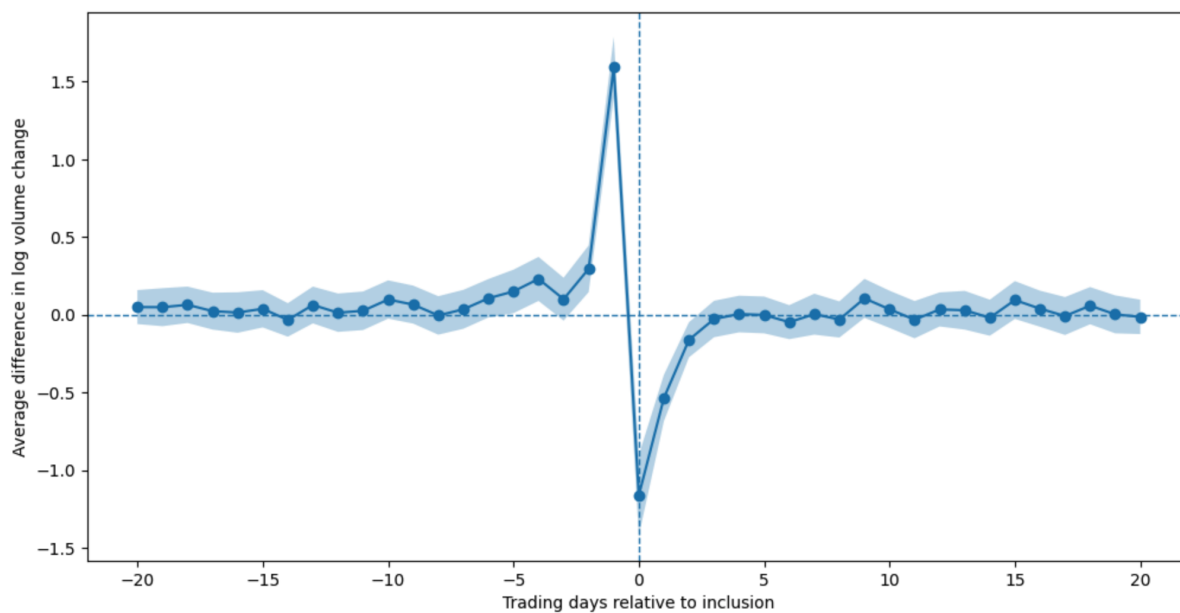


Figure 5: Log volume change;  $[-20, 20]$  window shown.

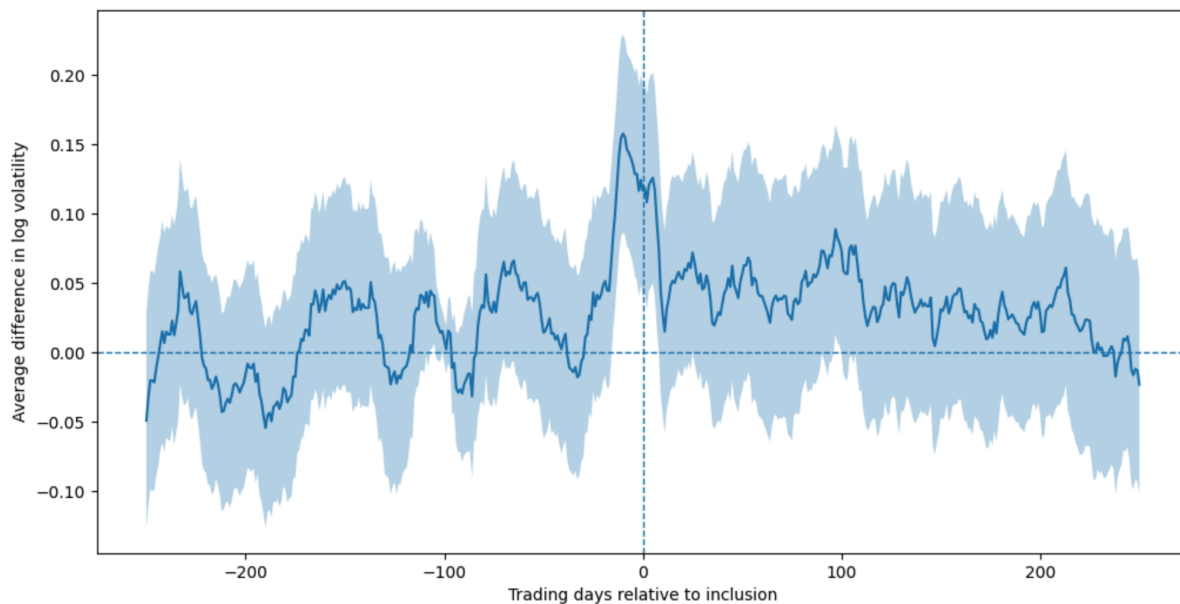


Figure 6: Log volatility;  $[-250, 250]$  window shown.

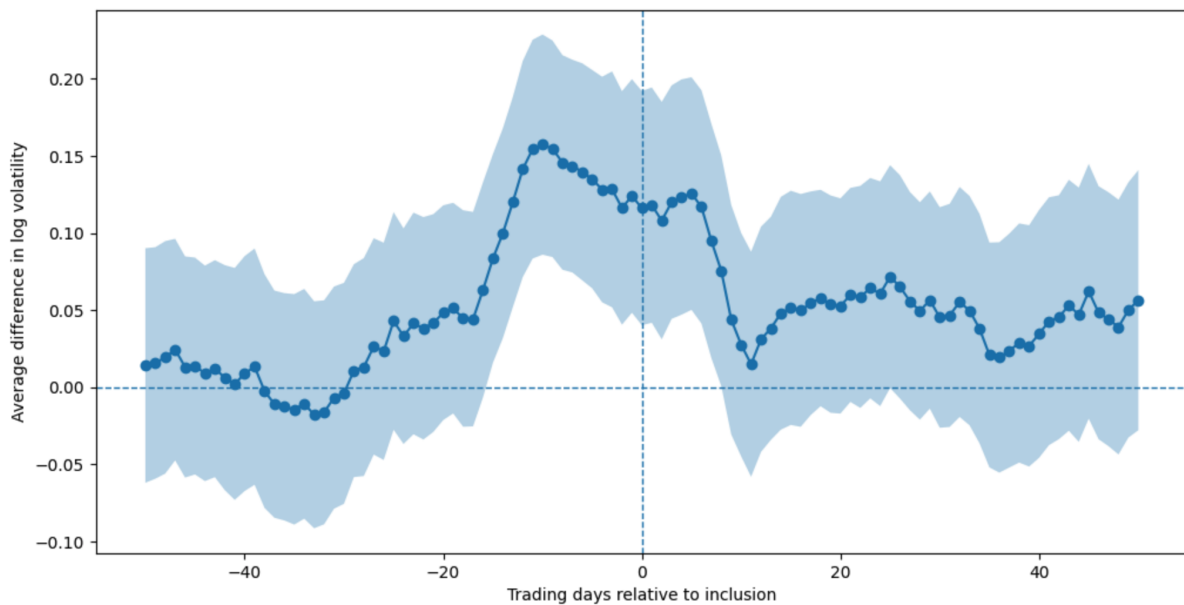


Figure 7: Log volatility;  $[-50, 50]$  window shown.

For Tables 1 and 2, \*, \*\* and \*\*\* indicate significance at the 10%, 5% and 1% levels respectively. Standard errors are heteroskedasticity-robust and clustered at the matched-pair level.

| $t$ | $\beta_t$ | Std. Error | $p$ -value |
|-----|-----------|------------|------------|
| -5  | 0.1497    | 0.0713     | 0.0358**   |
| -4  | 0.2301    | 0.0722     | 0.0014***  |
| -3  | 0.0998    | 0.0704     | 0.1562     |
| -2  | 0.2955    | 0.0766     | 0.0001***  |
| -1  | 1.5919    | 0.0993     | 0.0000***  |
| 0   | -1.1614   | 0.1321     | 0.0000***  |
| 1   | -0.5355   | 0.0759     | 0.0000***  |
| 2   | -0.1613   | 0.0576     | 0.0051***  |
| 3   | -0.0278   | 0.0598     | 0.6418     |
| 4   | 0.0043    | 0.0600     | 0.9420     |
| 5   | -0.0016   | 0.0603     | 0.9794     |

Table 1: Log volume change,  $\beta_t$  estimates around  $t = 0$ .

| $t$ | $\beta_t$ | Std. Error | $p$ -value |
|-----|-----------|------------|------------|
| -20 | 0.0483    | 0.0356     | 0.1744     |
| -19 | 0.0513    | 0.0349     | 0.1416     |
| -18 | 0.0445    | 0.0358     | 0.2141     |
| -17 | 0.0441    | 0.0355     | 0.2139     |
| -16 | 0.0631    | 0.0356     | 0.0764*    |
| -15 | 0.0835    | 0.0345     | 0.0157**   |
| -14 | 0.0998    | 0.0347     | 0.0041***  |
| -13 | 0.1200    | 0.0348     | 0.0006***  |
| -12 | 0.1414    | 0.0358     | 0.0001***  |
| -11 | 0.1544    | 0.0362     | 0.0000***  |
| -10 | 0.1574    | 0.0364     | 0.0000***  |
| -9  | 0.1547    | 0.0358     | 0.0000***  |
| -8  | 0.1458    | 0.0355     | 0.0000***  |
| -7  | 0.1434    | 0.0352     | 0.0000***  |
| -6  | 0.1397    | 0.0359     | 0.0001***  |
| -5  | 0.1349    | 0.0361     | 0.0002***  |
| -4  | 0.1282    | 0.0373     | 0.0006***  |
| -3  | 0.1283    | 0.0390     | 0.0010***  |
| -2  | 0.1163    | 0.0386     | 0.0026***  |
| -1  | 0.1240    | 0.0387     | 0.0013***  |
| 0   | 0.1162    | 0.0388     | 0.0027***  |
| 1   | 0.1181    | 0.0390     | 0.0024***  |
| 2   | 0.1081    | 0.0393     | 0.0060***  |
| 3   | 0.1201    | 0.0387     | 0.0019***  |
| 4   | 0.1234    | 0.0390     | 0.0015***  |
| 5   | 0.1256    | 0.0385     | 0.0011***  |
| 6   | 0.1170    | 0.0386     | 0.0025***  |
| 7   | 0.0949    | 0.0388     | 0.0144**   |
| 8   | 0.0754    | 0.0382     | 0.0482**   |
| 9   | 0.0438    | 0.0380     | 0.2493     |
| 10  | 0.0276    | 0.0370     | 0.4563     |
| 11  | 0.0148    | 0.0373     | 0.6903     |
| 12  | 0.0310    | 0.0373     | 0.4052     |
| 13  | 0.0382    | 0.0369     | 0.3011     |
| 14  | 0.0479    | 0.0385     | 0.2136     |
| 15  | 0.0514    | 0.0388     | 0.1846     |
| 16  | 0.0498    | 0.0385     | 0.1965     |
| 17  | 0.0544    | 0.0370     | 0.1417     |
| 18  | 0.0578    | 0.0358     | 0.1066     |
| 19  | 0.0539    | 0.0360     | 0.1342     |
| 20  | 0.0527    | 0.0356     | 0.1394     |

Table 2: Log volatility,  $\beta_t$  estimates around  $t = 0$ .

## 5 Discussion of Results

We begin with log returns. Figure 3 clearly shows no discernible effect, with both pre- and post-treatment trends barely deviating from zero. This provides strong evidence in support of Greenwood and Sammon (2025), who also used 100 trading days before inclusion as the reference period but measured cumulative abnormal returns instead. They found that increased liquidity and lower demand imbalances are the best explanations for the effect’s decrease over time; we would add to this the greater prevalence of technologically sophisticated arbitrageurs capturing well-known alpha.

Figures 4 and 5 corroborate the liquidity and arbitrage stories. Consistent pre-inclusion parallel trends give way starting at  $t = -5$  (and beyond all doubt by  $t = -2$ ) to massive spikes in trading volume: the estimate of  $\beta_{-1} = 1.5919$  reflects an average 391% ( $e^{1.5919} \approx 4.9131$ ) excess increase in volume just that day (it having already gone up significantly in previous days). By contrast, we have  $\beta_0 = -1.1614$ , giving a 69% decrease on the actual inclusion date that continues to a lesser extent on  $t = 1$  and  $t = 2$ . An institutional trading narrative makes sense here: index funds, derivatives traders etc. rebalance and arbitrage leading up to inclusion and gradually cool off (though still contribute to higher-than-normal volume—do not forget the cumulative, additive effects of the log changes!) in the days following. Overall, taking the point estimates from  $t = -5$  to  $t = 2$  at face value, the net change in trading volume is +66% (cf. Lin (2007), whose methodology is much better suited to provide this specific estimate and who gives a much more modest and believable  $\sim +10\%$ ; we are satisfied with the correct sign).

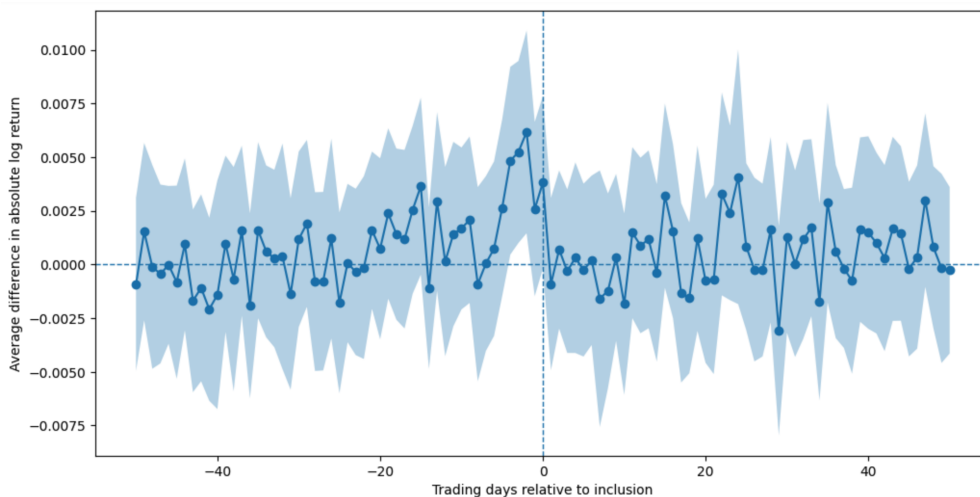


Figure 8:  $\beta_t$  estimates, absolute value of log return;  $[-50, 50]$  window shown.

Volatility is the most interesting of the three: most research (e.g., Hedge and McDermott (2003) and, again, Lin (2007)) suggests an increase, but some international evidence (e.g., Liu (2011)) points toward the opposite. Indeed, in Figures 6 and 7, the presence of a volatility spike shortly before the inclusion

date is clear in spite of serial correlation. However, Figure 8, though not a measure of volatility proper, gives us greater insight: we run the regression of the *absolute value of log return*, a sort of “one-day volatility.” The coefficients for  $t = -4$  ( $\beta_{-4} = 0.0048$ ,  $SE = 0.0022$ ,  $p = 0.0302$ ),  $t = -3$  ( $\beta_{-3} = 0.0052$ ,  $SE = 0.0022$ ,  $p = 0.0154$ ) and  $t = -2$  ( $\beta_{-2} = 0.0062$ ,  $SE = 0.0024$ ,  $p = 0.0103$ ) are significant, indicating turbulent trading on those dates; days  $t = -1$  and  $t = 0$  ( $p \approx 0.06$  for the latter) also have somewhat elevated values. Combining these insights, we conclude that volatility is not really higher except for this short period just before inclusion: the plots of Figures 6 and 7 begin to rise about ten days before and fall about ten days after this timespan, which lines up with how they were calculated. In conclusion, then, index inclusion seems to come with a positive volatility effect (very plausible for the institutional trading story), albeit one not nearly as significant as that of volume. An alternative avenue of analysis would be to consider the *implied volatility* from options markets, but we leave this for a future work.

While these results are informative, we obtained them by imposing a restrictive matching condition that, while largely fine for the S&P 500, is not practical for smaller indices or those in markets with less companies. Furthermore, we clustered standard errors at the matched-pair level, making it so that in order to improve estimates, we had to add two stocks’ worth of data at once. Ideally, we would want to use panel data of a large number of stocks over a long time (e.g., all companies listed on the NYSE or Nasdaq from 2000 to 2025, perhaps above a modest market capitalization filter), most of which are never treated and many of which might be treated multiple times, in order to expand coverage and reduce standard errors by sheer quantity of data. However, Goodman-Bacon (2021) showed that the conventional TWFE estimator distorts the average treatment effect in such circumstances: late-treated observations inappropriately act as controls for early-treated ones. Callaway and Sant’Anna (2021) offer a solution: they propose a novel bootstrap estimator to ascertain separate group-time average treatment effects and give ways to aggregate them. Sun and Abraham (2021) find problems for TWFE in and introduce an alternative interaction-weighted estimator for event studies specifically, while de Chaisemartin and d’Haultfoeulle (2023) focus on cases with multiple treatments, which is obviously relevant for index inclusion. We leave applications of these novel methods to larger datasets and different indices for future literature, but the opportunities are clearly abundant.

## 6 Conclusion

In this paper, we implemented a matched, event-study differences-in-differences design to study the S&P 500 index inclusion effect from 2007 to 2023, finding no trading price abnormalities but noticeable effects on volatility and especially volume. We reviewed the current literature and proposed future research directions in the context of pertinent econometric developments.

*Number one rule of Wall Street: nobody—I don’t care if you’re Warren Buffett or if you’re Jimmy Buffett—nobody knows if a stock is gonna go up, down, sideways or in [\*] circles. —Mark Hanna (The Wolf of Wall Street)*

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